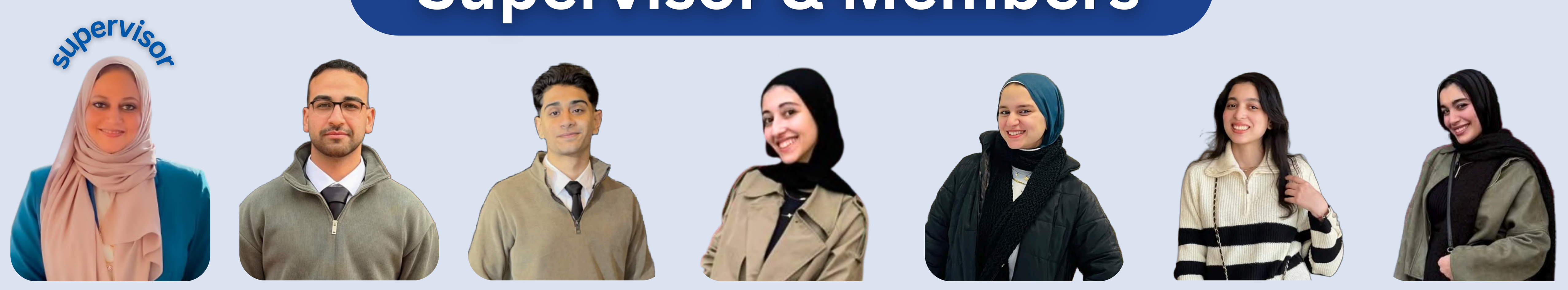
AI-Assisted 3D Brain Tumor Surgical Planning with Risk-Aware Trajectory Optimization

"Pre-Surgery Smart Planning System Using AI"

Supervisor & Members



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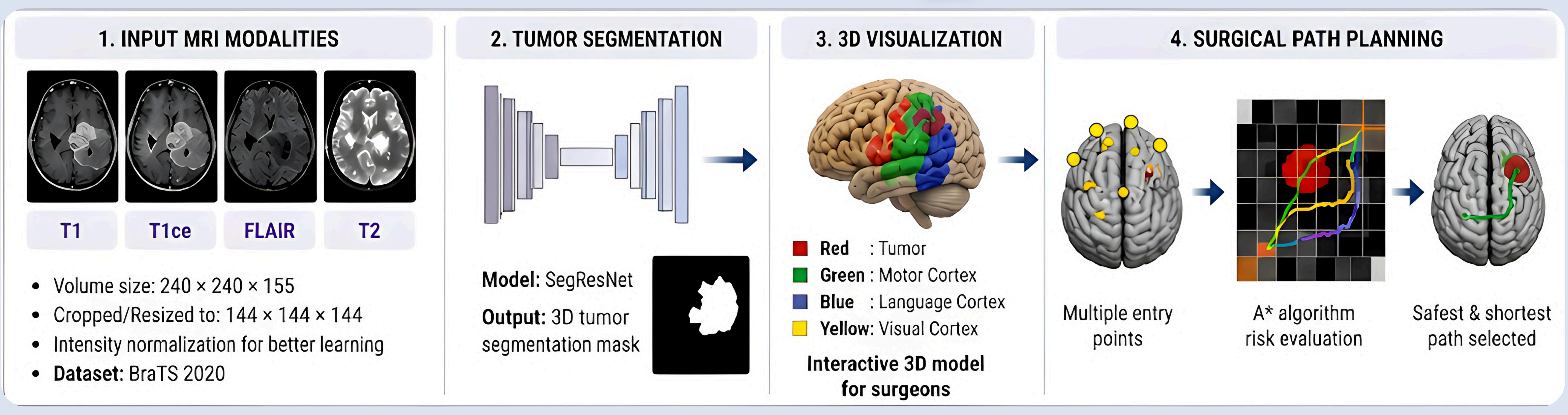
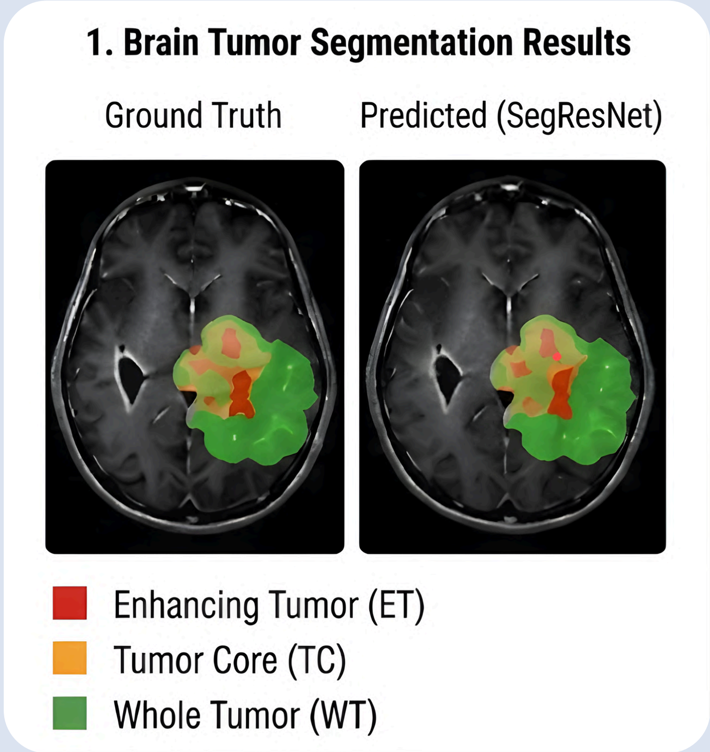
Abstract

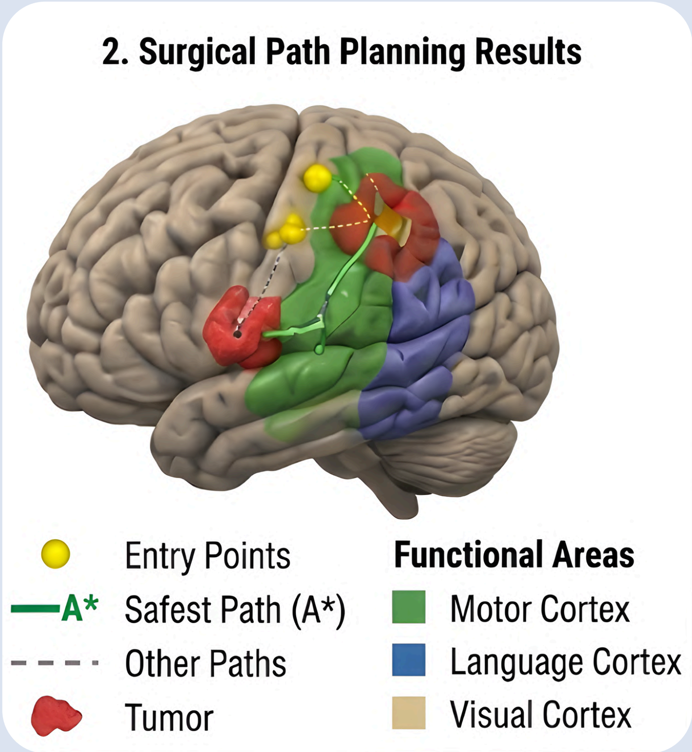
- AI-powered system for smart pre-surgical planning in brain tumor removal.
- Detects and segments tumor from multi-modal MRI using **SegResNet**.
- **3D visualization** of tumor with functional areas (motor, visual, language cortex).
- **A* algorithm** evaluates multiple entry points and recommends the safest,
- shortest surgical path avoiding critical brain regions.
- Results demonstrate **~%89** segmentation accuracy and significant reduction in surgical planning time, improving patient outcomes and procedural safety.

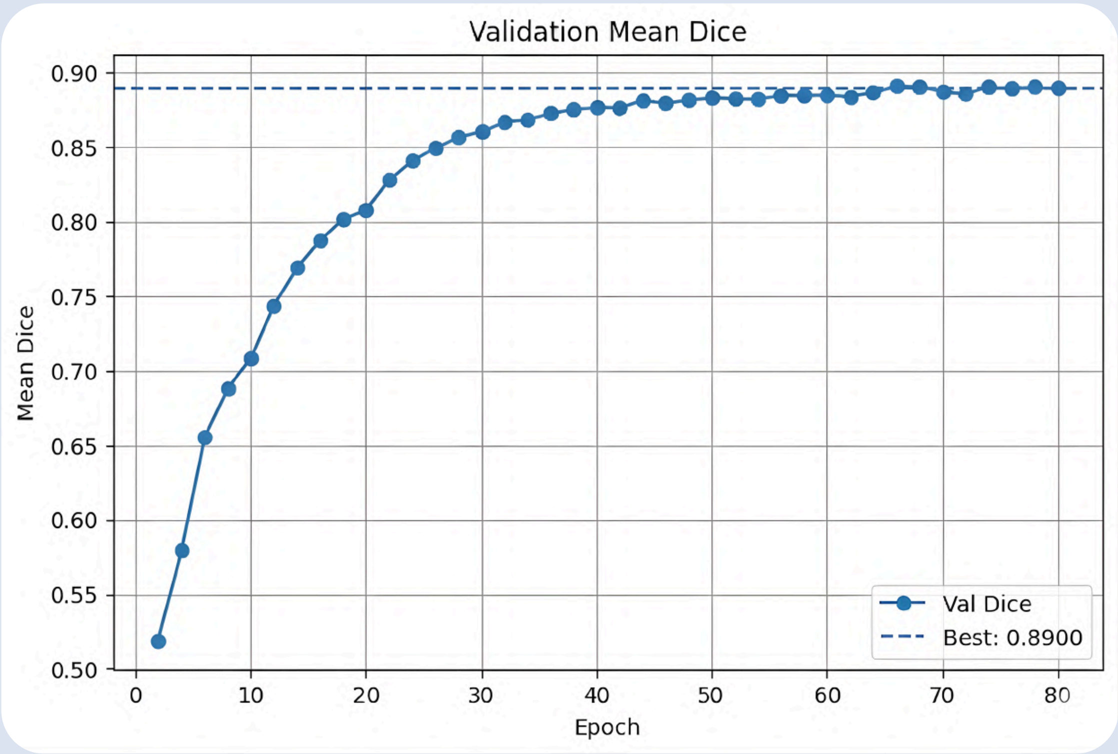
Methedology



Results







Discussion

High Segmentation Accuracy

The SegResNet model achieved ~89% Dice score, showing strong agreement with ground truth annotations.

Functional Awareness

Overlay of functional areas with tumor helps identify potential intersections and surgical risks.

Optimized Surgical Path

A* algorithm effectively finds the safest and shortest path while avoiding critical brain regions.

Clinical Relevance

Interactive 3D visualization provides intuitive insights, assisting surgeons in better decision making.

Conclusion

- Developed an end-to-end AI-based pre-surgery planning system.
- Accurate tumor detection and segmentation using SegResNet **(~89% Dice).**
- Integrated functional brain mapping for risk assessment.
- **A* based path** planning generates the safest and shortest trajectory.
- System provides interactive **3D visualization** for better surgical planning.

Future work

- Integrate **patient-specific DTI tractography** for white matter analysis.
- Incorporate real **fMRI** for more accurate functional mapping.
- Extend to **real-time surgical navigation** and intraoperative guidance.
- Improve performance using **transformer-based** models and larger datasets.
- Clinical validation through trials with **neurosurgeons**.

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GitHub